Goal and research questions

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We estimate how much each known COSMIC signature (for example SBS4 tobacco, SBS7 UV) contributed to a tumor from its 96-context mutation counts, compare four methods, and measure when estimates can be trusted. Outputs are research estimates, not diagnoses.

RQ1. How accurately do NNLS, elastic-net NNLS, a maximum-likelihood fit and a small neural network recover known exposures? RQ2. At what mutation count does accuracy break down? RQ3. Does performance hold under a cohort shift? RQ4. Is SBS4 weight higher in smokers with lung adenocarcinoma?

Data, methods and evaluation

Data. COSMIC SBS v3.4 signatures (exome-normalized, per-cancer-type subset, artifacts excluded). TCGA open-access MAFs filtered to MuTect2 calls, plus clinical data, for LUAD, SKCM and COAD. PCAWG only if an open download is confirmed in week 1.

Ground truth. Simulated tumors only. Patients are split 60/20/20 by TCGA patient ID before the simulator is calibrated on training patients.

Methods. B0 cohort-mean baseline, M0 NNLS, M1 elastic-net NNLS, M2 maximum likelihood (EM) with bootstrap intervals, M3 small MLP.

Experiments. E0 pipeline check, E1 method comparison, E2 accuracy vs mutation count (25 to 5,000), E3 unseen signatures, E5 SBS4 vs smoking with negative controls. Optional: E4 cohort shift.

Metrics. Exposure cosine, MAE, presence F1 and PR-AUC, confusion matrix, breakdown point N* (median cosine ≥ 0.90); paired Wilcoxon with Holm correction. Pilot: NNLS needed 500 to 1,000 mutations to reach 0.90.

Schedule and owners

Hugo: data. Bao Tran: simulator and classical methods. Anas: neural network and evaluation.

Week 1: data matrices; simulator, B0, M0, M1; metrics code. Week 2: splits and calibration; M2; train M3. Week 3: E1 to E3 and E5. Week 4: statistics, figures, report and presentation.

Minimum: B0, M0, M1 and M3 on LUAD and SKCM with E0 to E2 and E5. Main risk: TCGA download delays (fallback: simulation only).

Checklist

Question 1. Is the target clearly defined?
Answer: Yes. Each tumor's exposure vector over its cancer-type signature subset, known only for simulated tumors.

Question 2. Can source, patient or object identifiers leak across partitions?
Answer: No. We split by patient ID before simulator calibration and remove PCAWG donors also in TCGA.

Question 3. Is the positive class stated?
Answer: Yes. Signature weight of at least 5% for presence; current or reformed smokers for the smoking test.

Question 4. Would majority-class accuracy overstate success?
Answer: Yes. About 90% of labels are "absent", so we report F1, PR-AUC and cosine, and every method must beat B0.

Question 5. Can other classifiers reuse precisely the same split?
Answer: Yes. Splits and simulated test sets are saved once with fixed seeds and read by every method.

Question 6. Where will figures and intermediate results be stored?
Answer: In the private Git repository: results/, figures/ and splits/. Raw data is rebuilt from a download manifest.